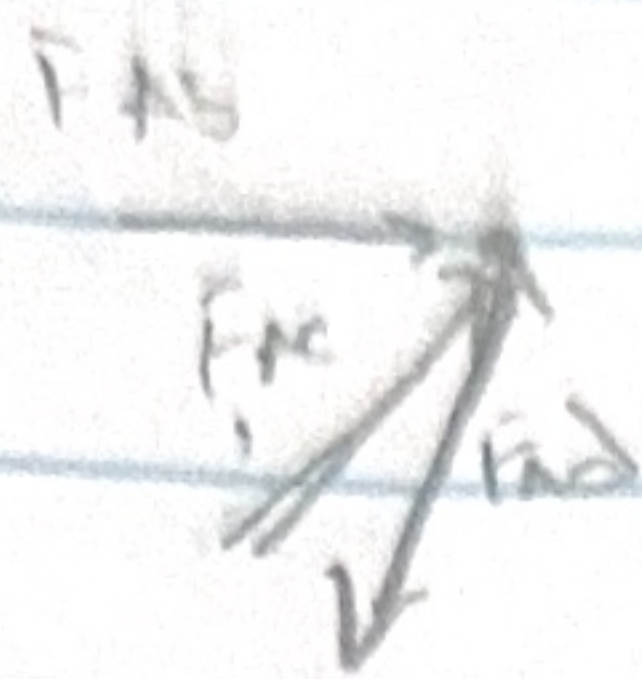


Joints.

Joint A



$$\sum F_x = 0$$

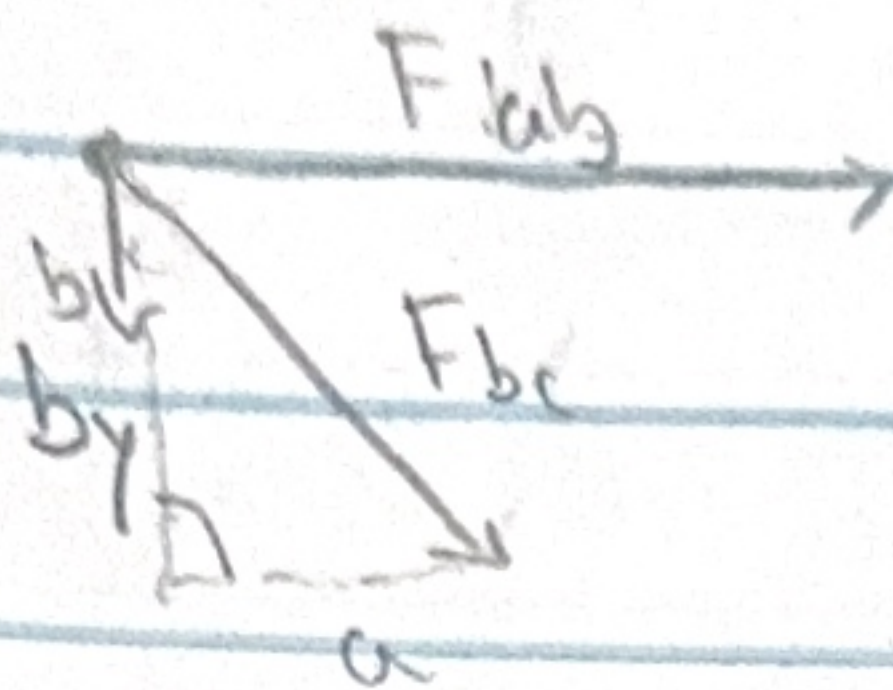
$$F_{AB} = \frac{P_a}{b}$$

$$\sum F_y = 0$$

$$F_{AC} - \frac{P \sqrt{a^2 + b^2}}{b} = 0$$

$$F_{AD} = \frac{P \sqrt{a^2 + b^2}}{b}$$

Joint B



$$\sum F_x = 0$$

$$F_{AB} - F_{BD} \cos(\theta) = 0$$

$$\frac{P_a}{b} -$$

$$\sum F_y = 0$$

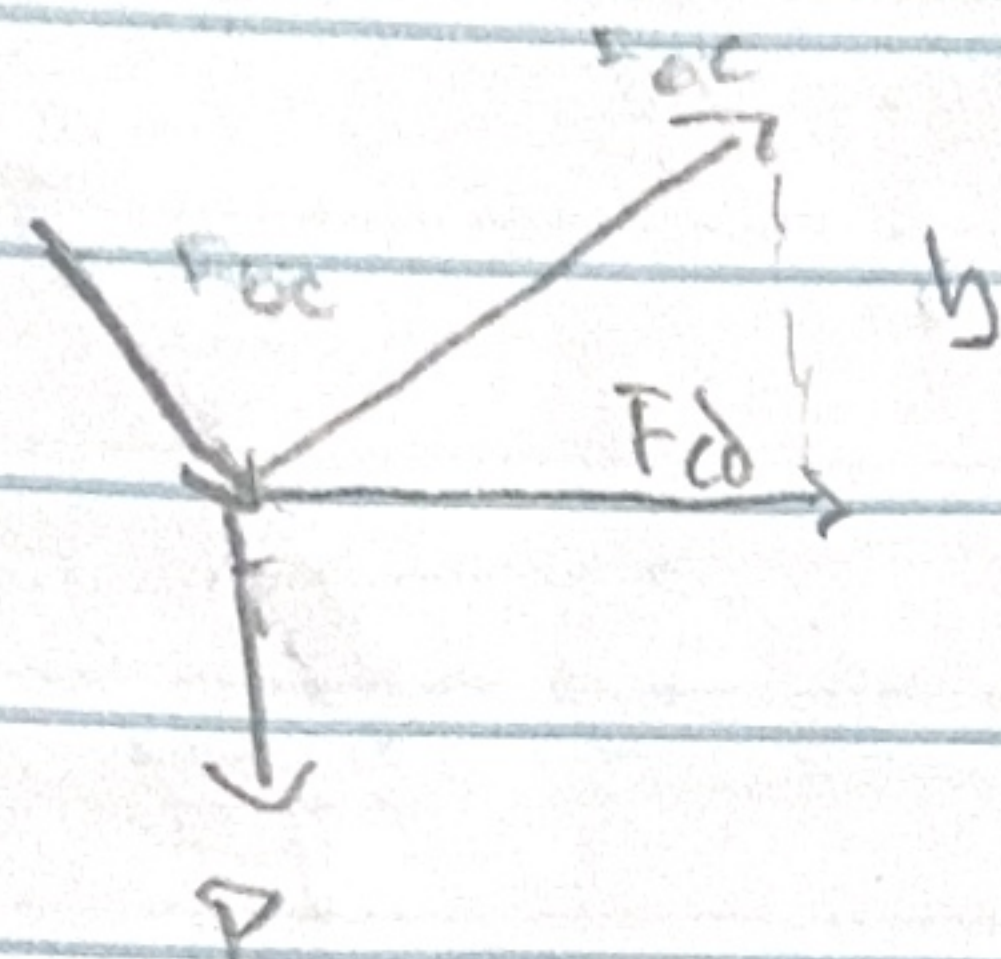
$$- \frac{b}{\sqrt{a^2 + b^2}} F_{BD} = B_y$$

$$-B_y - F_{BD} \sin(\theta) = 0 \quad F_{BD} = - \frac{B_y \sqrt{a^2 + b^2}}{b}$$

$$-B_y - F_{BD} \sin(\theta) = 0$$

$$+B_y$$

Joint C



$$\sum F_x = 0$$

$$+F_{CD} + F_{CB} \cos(\theta) = 0$$

$$\sum F_y = 0$$

$$+F_{CB} \sin(\theta) - P + F_{CA} \sin(\theta) = 0$$

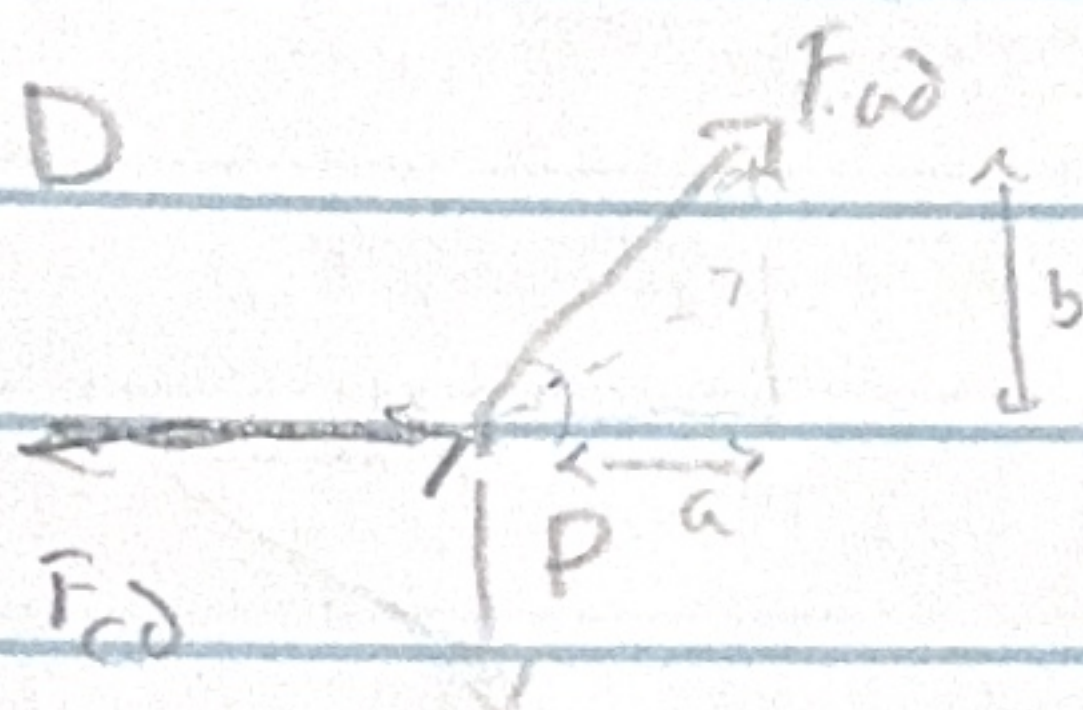
$$\left(\frac{b}{\sqrt{a^2 + b^2}} \right) \left(- \frac{P_a}{b} \right) + \left(\frac{b}{\sqrt{a^2 + b^2}} \right) F_{CA} = 0$$

$$P + F_{CA} \left(\frac{b}{\sqrt{a^2 + b^2}} \right) + \left(- \frac{P_a \sqrt{a^2 + b^2}}{b} \right) \left(\frac{b}{\sqrt{a^2 + b^2}} \right) = 0$$

$$P + F_{CA} \left(\frac{b}{\sqrt{a^2 + b^2}} \right) - P = 0$$

$$F_{CA} = 0$$

Joint D



$$\sum F_x = 0$$

$$F_{AD} \cos(\theta) - F_{CD} = 0$$

$$F_{CD} - F_{AD} \left(\frac{a}{\sqrt{a^2 + b^2}} \right) = 0$$

$$F_{CD} = - \frac{P_a}{b}$$

$$\sum F_y = 0$$

$$-P + F_{AD} \sin(\theta) = 0$$

$$F_{AD} \sin(\theta) = P$$

$$F_{AD} \frac{b}{\sqrt{a^2 + b^2}} = P$$

$$F_{AD} \frac{b}{\sqrt{a^2 + b^2}} = P$$

$$F_{AD} = \frac{P \sqrt{a^2 + b^2}}{b}$$

Symbolically

$$F_{ab} = D_0 / b \quad (+)$$

$$F_{bc} = P \sqrt{a^2 + b^2} / b \quad (c)$$

$$F_{ac} = 0$$

$$F_{cd} = P a / b \quad (t)$$

$$F_{ad} = \frac{P a^2 + b^2}{b} \quad (c)$$

Numerically

$$F_{ab} = 25 \text{ kN} \cdot \frac{0.4}{0.3} = 33.33 \text{ kN}$$

$$F_{bc} = 25 \text{ kN} \cdot \frac{0.5}{0.3} = 41.67 \text{ kN}$$

$$F_{ac} = 0 \text{ kN}$$

$$F_{cd} = 0.4 / 0.3 \cdot 25 \text{ kN} = 33.33 \text{ kN}$$

$$F_{ad} = 0.5 / 0.3 \cdot 25 \text{ kN} = 41.67 \text{ kN}$$